Star Cores: results of the first runs

Every new shape given the star core treatment, five ways of building on the spikes, and the old methods



DICTO · study results of 30 October 2026 · for DICTO · not for publication

1. Conclusions

1. Polar spikes always fill. If each spike's tip is put at its face's polar point instead of above the face's centre, the layer between the star and its halo is filled exactly by one pyramid over each corner and one tetrahedron over each edge. This held for all 48 convex shapes tested, with no overlaps, for ×φ and ×2. It looks like a general theorem, close to Conway's "join" (to be checked in the prior-art search). It is the logical rule for group C.

2. DICTO's sloping bases close every convex shape tested. For the OCD Stella, the DBT and the OTT they fill the convex halo exactly. For the nine Queens' slices they close with no overlaps but end in a skin of their own, slightly inside the convex halo: a new outer solid for each slice.

3. Only three cores return. TI (#78), the GTI and the TI Star core have cores that come back every two turns. Twenty more shapes have stars that fill exactly but cores that don't return.

4. Building on the star: one clean step. The first star always builds cleanly for the shapes whose own pyramids make them up. A second star, spiking the first star's triangles, clashes in the valleys under both the centre rule and the local rule. With DICTO's shorter spikes in the valleys, the stars grow three steps with no clash at all (all 75 runs finished so far reached the third step but one; the 21 runs on the largest Queens' slices are still going). Each spike is shortened just enough to clear its neighbours. The ridges keep full spikes, the valleys flatten, and the growth slows step by step. The most symmetric shapes freeze: at the third step most spikes of the FPD, the Palintriacontahedron and the Lion lie flat and the rest are too short to add any measurable volume (the OG adds 0.3 per cent), so the star stops growing. These are finished stars. The less symmetric pieces keep growing most: Lahdeli, Samson, Chucky, the Triskelion Tetrahedron and Fang.

5. The old methods confirm the network. The duals give the known pairs (GTI with FPD and the Palintriacontahedron, OG with the Hecatodihedron, the Star Halo with the TI Star core). The tips of Tweedledum's first stellation make a regular dodecahedron. Nothing else came out with all edges equal or all faces regular.

2. Every shape (×φ)

Group: **A** star fills and the core returns · **B** star fills, core does not return · **C** star inside its halo, layer not filled by corner and edge pieces · **D** star sticks out of its halo. Building columns: the volume each step adds (× the step before); ✗ marks a step with clashes. Valley: the volume each step adds, with the spikes shortened in the valleys; 'running' marks the runs still going.

D	Shape	Group	Polar spikes	Sloping bases	Build on star, centre rule	Build on star, local rule	Valley rule	Dual is
D1	skewed rhombic dodecahedron	D	✓ fills	–	1.618 · 1.618✗	1.618 · 1.472✗	1.618 · 1.128 · 1.077	
D2	skewed elongated dodecahedron ED F16	D	✓ fills	–	1.618 · 1.671✗	1.618 · 1.472✗	1.618 · 1.115 · 1.015	
D3	skewed elongated dodecahedron ED F18	D	✓ fills	–	1.618 · 1.642✗	1.618 · 1.472✗	1.618 · 1.090 · 1.084	
D4	Jewel (EKP windows; first called Dragon Jewel)	C	–	–	1.618✗	1.618✗	1.412 · 1.361 · 1.187	
D6	Dodeca-13 Wedge	D	✓ fills	–	1.618 · 1.618✗	1.618 · 1.564✗	1.618 · 1.344 · 1.285	
D7	Dodeca-13 Needle	D	✓ fills	–	1.618 · 1.618 · 1.830✗	1.618 · 1.708✗	1.618 · 1.177 · 1.279	
D22	Truncated-Icosahedral Hecatodihedron	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.574✗	1.618 · 1.041 · 1.002	Ogdoecontahedron (OG)
D28	Ogdoecontahedron (OG)	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.486✗	1.618 · 1.031 · 1.003	Truncated-Icosahedral Hecatodihedron
D29	OCD Stella	C	✓ fills	✓ fills halo	1.618 · 1.618✗	1.618 · 1.635✗	1.618 · 1.034 · 1.000	
D30	Metaxyhexecontahedron	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	1.618 · 1.279 · 1.012	
D31	Palintriacontahedron	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	1.618 · 1.281 · 1.000	GTI (golden truncated icosahedron)
D37	Belted Triacontahedron (DBT)	C	✓ fills	✓ fills halo	1.618 · 1.618✗	1.618 · 1.517✗	1.618 · 1.301 · 1.000	
D38	OTT (Octahedrally Truncated Triacontahedron)	C	✓ fills	✓ fills halo	1.618 · 1.618✗	1.618 · 1.658✗	1.618 · 1.015 · 1.000	
D40	GTI (golden truncated icosahedron)	A	✓ fills	–	1.618 · 1.618✗	1.618 · 1.503✗	1.618 · 1.476 · 1.094	FPD (flat pentakis dodecahedron)
D41	FPD (flat pentakis dodecahedron)	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	1.618 · 1.281 · 1.000	GTI (golden truncated icosahedron)
D45	KRE	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.491✗	1.618 · 1.041 · 1.002	
D46	PHE	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.526✗	1.618 · 1.130 · 1.001	
D47	TPP	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.635✗	running	
D48	DPHP	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.504✗	1.618 · 1.096	
D49	TPH	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.657✗	1.618 · 1.192 · 1.000	
D50	TQH	B	✓ fills	–	1.618 · 1.711✗	1.618 · 1.636✗	1.618 · 1.095 · 1.000	
D51	TQP	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.571✗	1.618 · 1.041 · 1.002	
D52	TQHD	B	✓ fills	–	1.618 · 1.658✗	1.618 · 1.572✗	running	
D69	Lotus Jewel Axes	C	–	–	1.618✗	1.618✗	1.447 · 1.191 · 1.213	
D70	Dodecaxes	C	–	–	1.618✗	1.618✗	1.095 · 1.108 ·	

D	Shape	Group	Polar spikes	Sloping bases	Build on star, centre rule	Build on star, local rule	Valley rule	Dual is
							1.147	
D71	Samson	D	✓ fills	–	1.618 · 1.618✗	1.618 · 1.577✗	1.618 · 1.116 · 1.064	
D76	Sonsam	D	✓ fills	–	1.618 · 1.645✗	1.618 · 1.546✗	1.618 · 1.081 · 1.040	
D77	Lahdeli	D	✓ fills	–	1.618 · 1.618✗	1.618 · 1.616✗	1.618 · 1.134 · 1.074	
D78	Ian Dury	D	✓ fills	–	1.618 · 1.636✗	1.618 · 1.527✗	1.618 · 1.148 · 1.071	
D79	Blockhead	D	✓ fills	–	1.618 · 1.618✗	1.618 · 1.551✗	1.618 · 1.165 · 1.167	
D80	Chucky	D	✓ fills	–	1.618 · 1.618✗	1.618 · 1.493✗	1.618 · 1.266 · 1.257	
D81	Fang	D	✓ fills	–	1.618 · 1.618✗	1.618 · 1.587✗	1.618 · 1.344 · 1.158	
D82	Triskelion Tetrahedron	D	✓ fills	–	1.618 · 1.618✗	1.618 · 1.493✗	1.618 · 1.063 · 1.081	
D83	Keyaxes	C	–	–	1.618✗	1.618✗	1.239 · 1.191 · 1.151	
D84	Ganesha	D	✓ fills	–	1.618 · 1.656✗	1.618 · 1.503✗	1.618 · 1.112 · 1.089	
D86	Kagemusha	C	–	–	1.618✗	1.618✗	1.458 · 1.504 · 1.189	
D87	Lion	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	1.618 · 1.169 · 1.000	
D88	Jabberwock	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	1.618 · 1.077 · 1.000	
D89	Tweedledum	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	1.618 · 1.235 · 1.000	
D90	Tweedledee	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	1.618 · 1.269 · 1.118	
D91	Mad Hatter	C	✓ fills	closed, own skin	1.618 · 1.648✗	1.618 · 1.638✗	running	
D92	March Hare	C	✓ fills	closed, own skin	1.618 · 1.639✗	1.618 · 1.643✗	running	
D93	Jubjub	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	1.618 · 1.251 · 1.063	
D94	Alice	C	✓ fills	closed, own skin	1.618 · 1.648✗	1.618 · 1.632✗	running	
D95	White Rabbit	C	✓ fills	closed, own skin	1.618 · 1.646✗	1.618 · 1.645✗	running	
D96	Haigha	C	✓ fills	closed, own skin	1.618 · 1.618✗	1.618 · 1.499✗	running	
D97	Hatta	C	✓ fills	closed, own skin	1.618 · 1.621✗	1.618 · 1.497✗	running	
D98	Red King	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.512✗	running	
D99	Walrus	C	✓ fills	closed, own skin	1.618 · 1.621✗	1.618 · 1.489✗	running	
D100	Red Knight	C	✓ fills	closed, own skin	1.618 · 1.618✗	1.618 · 1.497✗	running	
D101	White Knight	C	✓ fills	closed, own skin	1.618 · 1.618✗	1.618 · 1.476✗	running	

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D103	Golden TI Star	D	–	–	1.618✗	1.618✗	running	
D104	Star Halo	B	✓ fills	–	1.618 · 1.618✗	1.618 · 1.708✗	running	TI Star core
D105	TI Star core	A	✓ fills	–	1.618 · 1.618✗	1.618 · 1.510✗	running	Star Halo

Studies 2026-10-30-star-cores (star_core.py, star_build.py, polar_spikes.py, sloping_bases.py, star_valley.py, group2_diag.py) and 2026-10-30-old-methods (old_methods.py), run on dicto-big-brother; numerical (float64), overlaps by the separating-axis test. Exact confirmation follows for the results that go into papers.